Project Kick-off

Comprehensive Design of Computer Science and Software Engineering

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Today’s four sessions

01 Project requirements

02 Student introductions

03 2025 project showcase & this year’s topics

04 WeChat groups and team formation

SESSION 01

Project requirements

A working system with clear engineering evidence

Proposal presentation and final defense

Proposal presentation

What we plan to build
Why it matters
How we will build and test it

Final defense

What we actually built
How the system works
What the evidence shows

The proposal reviews the plan. The final reviews the completed system.

Proposal: problem, users, and scope

Problem	Background, target users, and a specific user need.
Related work	Existing approaches and the gap your project addresses.
Core scope	2-3 user stories and one complete main workflow.

Explain whose problem you will solve.

Proposal: system design and work plan

Technology	Technology stack, data sources, and the role of AI.
Risks and tests	Main risks, mitigation, and planned verification.
Team plan	Member roles and the plan of work.

Use a clear architecture or workflow diagram to explain the plan.

Proposal presentation: delivery requirements

Presentation and Q&A

Present one coherent team plan.
Explain scope and feasibility.
Answer questions about choices.

Submission

One proposal slide per team.
Submit the slides as a PDF.

Present in English. A finished system is not required at the proposal stage.

Final: working system and demonstration

Runnable system	A usable interface, backend logic, and connected data.
Core workflow	Show a real task from input to a useful result.
Demonstration	Live or recorded system demo of at least two minutes.
Failure handling	Show how the system handles invalid input or failure.

Present in English. Demonstrate the system you completed and explain its current limitations.

Final: engineering and AI evidence

Design	Architecture, module responsibilities, and key decisions.
Verification	Core tests, failure cases, and repeatable checks.
AI evaluation	Evaluate AI behavior or the AI-assisted workflow.
AI use record	Tools, tasks, human changes, checks, and remaining risks.

Support your claims with results that another person can inspect.

Final technical report

Problem and context	Target users, requirements, scope, and related work.
System and methods	Architecture, implementation, AI role, and design choices.
Results and limitations	Tests, evaluation, failure cases, and known limitations.
Team and sources	Member contributions, AI usage statement, and references.

Write in English. Use figures and tables to explain the completed work.

Final submission and defense

Software

Code repository
README and run steps

Presentation slides & Demo video

Final slides as PDF
System demo

Report

Final report

Submit one team package.

Course assessment

10%

Attendance and participation

40%

Full-stack software labs

50%

Final project

Project milestones

年月、周次 / 日期 / 星期	2026年																			2027年						
	9月				10月					11月				12月				1月			2月					
星期	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	寒假					
星期一	31	7	14	21	28	5	12	19	26	2	9	16	23	30	7	14	21	28	4	11	18	25	1	8	15	22
星期二	1	8	15	22	29	6	13	20	27	3	10	17	24	1	8	15	22	29	5	12	19	26	2	9	16	23
星期三	2	9	16	23	30	7	14	21	28	4	11	18	25	2	9	16	23	30	6	13	20	27	3	10	17	24
星期四	3	10	17	24	1	8	15	22	29	5	12	19	26	3	10	17	24	31	7	14	21	28	4	11	18	25
星期五	4	11	18	25	2	9	16	23	30	6	13	20	27	4	11	18	25	1	8	15	22	29	5	12	19	26
星期六	5	12	19	26	3	10	17	24	31	7	14	21	28	5	12	19	26	2	9	16	23	30	6	13	20	27
星期日	6	13	20	27	4	11	18	25	1	8	15	22	29	6	13	20	27	3	10	17	24	31	7	14	21	28
	教学 18 周																		考试2周		寒假 6 周					

Teamwork and shared responsibility

Team size	3-4 students per team, following Lecture 01.
Shared work	One codebase and one coordinated submission.
Contribution	Record implementation, design, testing, and review work.
Responsibility	Everyone should explain their work and the system.

Some teams will collaborate across schools.

设计学院参与介绍

Questions about the project

What would you like to clarify before forming a team?

Scope, deliverables, AI use, or collaboration

SESSION 02

Student introductions

Programming experience and AI use

School of Future Technology

About me	Name, and programming experience
My AI tool	Which tool do I use, and for what development task?
One example	What did AI produce? What did I check or change?

No prior AI experience? Share what you would like to try.

设计学院同学自我介绍

SESSION 03

2025 project showcase & Topics for this year’s project

Working systems and the decisions behind them

What to notice in each project

User experience

Who uses it?
What task becomes easier?
Where does the user act?

Software

What actually runs?
How do the parts connect?
What does AI contribute?

Evidence

What did the team test?
What limitations remain?
What would you improve?

Project options for this year 2026

01 Mobile GUI Task Agent

02 AI Personal Finance Assistant

03 AI Travel Planning and Action Assistant

04 AI Meeting-to-Action Assistant

X Open Innovation Project

Choose one project and a manageable core workflow.

Project 1: Mobile GUI Task Agent

Mobile interaction and cross-app automation

Understand a user request, read the phone screen, and operate existing apps to complete a task.

Example request

“Open the phone’s built-in scanner, read this QR code, open the link, and email it to the specified contact.”

Understand the request. Observe the screen. Act. Check the result.

Project 1: core modules (1/2)

Module	Required function	AI / integration
Task understanding	Extract goals, apps, contacts, and content. Ask when details are missing.	LLM parsing Structured tasks
Screen understanding	Identify text, controls, input fields, and the current page state.	Screenshot / OCR Vision model / UI tree
Action execution	Open apps, tap, swipe, type, and navigate back.	Platform automation GUI action tools

Reference requirements for the core workflow.

Project 1: core modules (2/2)

Module	Required function	AI / integration
Cross-app workflow	Pass the scanned link from the built-in scanner to the browser and email app.	App switching Task state management
Verification and recovery	Check success. Handle delays, pop-ups, missing targets, and retries.	State checks Replanning / retries
User control	Show progress. Support pause, cancel, and confirmation when needed.	Execution log Human takeover

Implementation may combine model reasoning with deterministic checks.

Project 1: scope and demonstration

Baseline scope

Choose one phone platform, e.g. Android.
Use 2 apps and a few tasks.
Adapt actions to the current screen.
Keep the action trace and result.

Finish the example of scanner+QR code.

Optional extensions

Voice input
Learning from a demonstration
Multilingual commands
…

Demo: scan → open link → email contact; recover from a pop-up or failed action.

Project 2: AI Personal Finance Assistant

Personal finance and explainable decision support

Organize financial records and help users understand spending or assets, compare scenarios, and track goals.

Example request

“Use my last three months of spending to plan how I could save RMB 6,000 in six months.”

Import records. Calculate the overview. Compare scenarios. Track progress.

Project 2: core modules (1/2)

Module	Required function	AI / integration
Data import	Import transactions, assets, or holdings. Allow edits and detect duplicates.	CSV / Excel parsing Optional OCR
Record organization	Classify income, spending, and transfers. Let users correct categories.	Rules and LLM User corrections
Financial overview	Show spending trends, budget use, or asset allocation. Link to source records.	Database Calculations / charts

Reference requirements for the core workflow.

Project 2: core modules (2/2)

Module	Required function	AI / integration
Goal and scenario planning	Calculate progress and compare changes in amount, deadline, or budget.	Calculation tools Scenario simulation
Grounded Q&A	Answer from imported data. Explain calculations and missing information.	Data retrieval LLM tool use
Monitoring and reminders	Flag user-defined budget overruns, goal gaps, or asset concentration.	Rule checks Notifications

Implementation may combine model reasoning with deterministic checks.

Project 2: scope and demonstration

Baseline scope

Choose spending or asset tracking.
Use imported or labeled sample data.
Use code for amounts and ratios.
State assumptions and data sources.

Optional extensions

Receipt recognition
Recurring-payment detection
Multiple currencies
Historical portfolio review

Demo: trace one analysis to the records, then change a planning assumption.

Project 3: AI Travel Planning and Action Assistant

Travel planning and multi-tool agents

Turn travel preferences into a checked, editable itinerary and carry a confirmed plan into another app.

Example request

“Plan a two-day trip from Guangzhou under RMB 1,500 with less walking. Open the route and add the itinerary to my calendar.”

Collect preferences. Check information. Revise the plan. Execute an action.

Project 3: core modules (1/2)

Module	Required function	AI / integration
Preference understanding	Extract dates, origin, budget, group size, and preferences. Ask for missing details.	LLM dialogue Structured constraints
Information gathering	Find places, transport, and opening hours. Keep sources and query times.	Maps / search Web or tool access
Plan generation	Create an itinerary with places, time slots, and estimated costs.	LLM planning Structured itinerary

Reference requirements for the core workflow.

Project 3: core modules (2/2)

Module	Required function	AI / integration
Constraint checking	Check time conflicts, budget, and routes against the request.	Map data Calculation / rules
Interactive revision	Revise the plan when preferences change and explain the differences.	Conversation state Plan comparison
Action execution	Open a confirmed route in maps or create calendar events.	App interfaces Shortcuts / GUI tools

Implementation may combine model reasoning with deterministic checks.

Project 3: scope and demonstration

Baseline scope

Limit the region or trip type.
Generate, edit, and save a plan.
Complete a real map or calendar action.
Separate quoted prices from estimates.

Optional extensions

Replanning after weather changes
Preferences from several travelers
Travel screenshot recognition
Accessibility preferences

Demo: change a constraint, revise the itinerary, and carry out one app action.

Project 4: AI Meeting-to-Action Assistant

Speech understanding and workflow automation

Turn meeting records into reviewable decisions, tasks, and calendar events, then execute confirmed actions.

Example request

“Extract everyone’s tasks, add next Wednesday’s meeting to the calendar, and send the confirmed plan to the project group.”

Read or transcribe. Extract tasks. Review details. Create and verify actions.

Project 4: core modules (1/2)

Module	Required function	AI / integration
Meeting input	Import a short recording or text notes. Show an editable transcript.	Speech recognition Text import
Decision and task extraction	Extract decisions, actions, owners, and deadlines.	LLM extraction Structured output
Evidence linking	Link each extracted item to the source text or recording time.	Text references Timestamps

Reference requirements for the core workflow.

Project 4: core modules (2/2)

Module	Required function	AI / integration
Human review	Correct names, tasks, and dates. Mark missing information for review.	Editing interface Ambiguity checks
Action execution	Create confirmed calendar events or tasks, or send a specified message.	Calendar / task APIs App automation
Status tracking	Show success, failure, or pending review. Detect possible duplicate actions.	Execution state Duplicate checks

Implementation may combine model reasoning with deterministic checks.

Project 4: scope and demonstration

Baseline scope

Support short meetings or text notes.
Let users correct owners and dates.
Connect one real execution channel.
Detect possible duplicate actions.

Optional extensions

WeChat message delivery
Speaker identification
Progress review across meetings
Task reminders

Demo: trace a task to the meeting record and create it in a real application.

Project X: Open Innovation Project

Possible directions

Desktop task agents
Household assistants
Accessibility assistants
Shopping comparison agents

Your proposal explains

User and concrete task
Data, tools, and AI role
Success checks and recovery
Feasibility and team plan

The same proposal and final requirements apply.

Requirements shared by all project options

Runnable workflow	Connect the interface, data, AI, and tools.
User control	Let users inspect results, correct inputs, and intervene.
Verification	Show normal use, an abnormal case, and its handling.
Delivery	Follow the existing proposal and final requirements.

Understand the request, perform the task, and verify the result.

SESSION 04

WeChat groups and team formation

Open discussion and team matching

Project communication on WeChat

Class project group

Announcements and shared materials
Questions and teammate matching
Display name: name + school / major

Team group

Create a group with your teammates
Agree on a contact person
Discuss a regular meeting time

Join through the group invitation shared in class.

Open discussion

Interests

What problems interest you?
What would you like to build?

Contribution

What can you offer?
What do you want to learn?

Working together

When can you meet?
How will you communicate?

Move around, meet classmates, and discuss possible teams.

Team information deadline

29 September 2026
23:59

Send your team information to the monitor.

The monitor collects all teams and forwards the list
to the instructor or teaching assistant.